

What is money?

The stock of assets that can be readily used to make transactions.

Fiat money:- Established as money by the govt decree, or fiat.

3 Functions:-

- Medium of Exchange
- Unit of Account
- Store of Value

"Liquidity of Money"

↳ The ease with which money is converted into other things - goods & services

$$\begin{aligned}\text{Money Supply} &= \text{Currency} + \text{Deposits} \\ (M_s) &= (C_u + D) \\ &= \left(\frac{C_u}{D} + 1 \right) \cdot D\end{aligned}$$

$$\begin{aligned}\text{High Powered Money} &= \text{Currency} + \text{Reserves} \\ \text{Reserve Money (H)} &= (C_u + R)\end{aligned}$$

$$\Rightarrow H = \left(\frac{C_u}{D} + \frac{R}{D} \right) \cdot D$$

$$\therefore M_s = \frac{\left(\frac{C_u}{D} + 1 \right)}{\left(\frac{C_u}{D} + \frac{R}{D} \right)} \cdot H$$

↳ m (Money multiplier)

$$M_s = f(H) ; f' \geq 0 \quad \text{[Ch. 16/Pg 347]}$$

$\Rightarrow M_s$ is a rising as a function of High Powered Money

$$m = \frac{M_s}{H} > 1 \quad M_s = mH ; m \geq 1$$

C/D $\rightarrow$ Currency - Deposit Ratio

R/D $\rightarrow$ Reserve - Deposit Ratio

Money Supply in India:

Narrow Money (M_1)

Govt. Securities / Govt. Bonds : Govts/bi-Secs

Govt. Edged Securities

PD's : Primary Dealers (Nomura)

MR. X $\rightarrow$ Sells Govts

RBI Buying the Govts $\rightarrow$ Injects Liquidity

$\hookrightarrow$ RBI issues a cheque

Mr. X receives the cheque

He goes to eg: Canara Bank

RBI credits the cheque money to the Reserve Account of Canara Bank.

$\hookrightarrow$ Reserves $\uparrow \rightarrow H \uparrow \rightarrow M_s \uparrow$

→ RBI sells "Giltts / G-Secs" ^{Uses (squeeze) Liquidity}
 MR. X Buying $\Rightarrow$ MR. X pays them through a cheque

RBI Debits from Reserve account of Canara Bank

Reserves $\downarrow \rightarrow H \downarrow \rightarrow M \downarrow$

* OMO : Open Market Operation

→ Suppose RBI buys G-Secs/Giltts worth of Rs. 100

MR. X sells G-Secs worth of Rs. 100

RBI issues a cheque in favour of Mr. X worth of Rs. 100

(Mandatory Reserve Requirement by the RBI)

$\hookrightarrow$ let 20% for Canara Bank & HDFC

Mr. X's Bank
Canara Bank

Asset
Rs. 100

Liability
Rs. 100

Rs. 20 Reserves with RBI

Rs. 80 $\rightarrow$ Can be used by the bank for loans

MR. Y (Takes loan Rs. 80 from Canara) & deposits it to HDFC

Asset
Rs. 80

Liability
Rs. 80

Rs. 16 Reserve

Rs. 64 Loans

$$100(1 + (0.8) + (0.8)^2 + \dots) = \frac{100}{1-0.8} = 500$$

Money being used for lending loans

∴ RBI maintains & controls money flow with the help of mandatory reserve requirement, ∴ by holding money in reserves as per requirement.

The Mechanism of Money Creation

→ Excess Reserves = [Actual Reserves - Required Reserves]

↳ Banks can lend loans only until you have excess reserves > 0 .

Cash Reserve Ratio (CRR)

(SLR) → Statutory Liquidity Ratio

↓

(For GLTs)

↳ 18% [Oct 2025]

Theory of Money Demand

Asset Market : Two Kinds of Assets

Money

Bonds

Money is ~~not~~ a non-interest bearing assets
Bonds are interest bearing assets

Question : How much money to hold?

and How much to keep in bonds?

$M^d \rightarrow$ Nominal Money Demand

Pg. 233

6th Nov
2025

B_p^t (Bond price)

6th Nov
2026

$B_p^{(t+1)}$

@Y $\rightarrow$ Interest rate

(Date of Maturity)
(DOM)

$$B_p^{t+1} = B_p^t + r \cdot B_p^t$$

(Maturity Value)
(known)

$$\bar{B}_p = (1+r) B_p^t$$

$$\Rightarrow B_p^t = \frac{\bar{B}_p}{(1+r)}$$

$$\text{So, } B_p^t \propto \frac{1}{r}$$

Real Money Demand

$$\frac{M^d}{P}$$

$$= L(Y, r) \quad \text{with } \frac{\partial L}{\partial Y} > 0 \quad \frac{\partial L}{\partial r} < 0$$

Indicates
purchasing
power.

$$\Rightarrow \frac{M^d}{P} = kY - hr \quad ; k > 0, h > 0$$

as Y rises Demand for Money $\uparrow$

Walras' Law $\div$ (SS $\rightarrow$ Supply)

$$\text{Total Money Demand} + \text{Total Bond Demand} + \text{SS} = 0$$

$$(M^d + B^d) = (M^s + B^s)$$

$$(M^d - M^s) = (B^s - B^d)$$

if $M^d - M^s > 0 \Rightarrow (B^s - B^d) > 0$

Excess demand
in the money
market

Excess supply in
the bond market.

→ If $M^s \downarrow \Rightarrow$ Excess demand in
money mkt.

Walras' Law



Excess supply in
bond mkt

$$\Rightarrow B_p \downarrow \quad \left. \vphantom{\Rightarrow B_p \downarrow} \right\} \because B_p \propto \frac{1}{r}$$

$$\Rightarrow r \uparrow$$

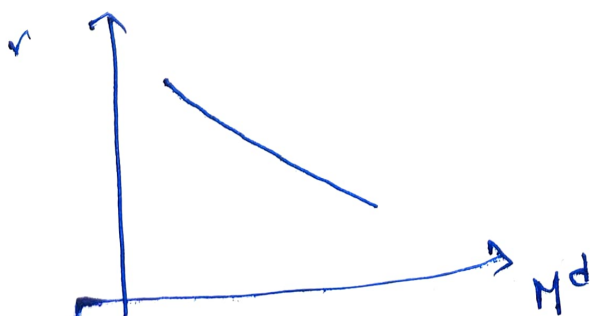
M^s is controlled by RBI

→ Bond Market is Mirror Image of
The money Market

$$\rightarrow M^d < M^s \rightarrow B_p \uparrow \text{ \& } r \downarrow$$

(or)

$$M^d > M^s \rightarrow B_p \downarrow \text{ \& } r \uparrow$$



Keynes in 1923

Three motives behind holding / demanding money

(i) Transactions demand for money

(ii) Speculative demand for money

↳ On Speculations of money gain/loss
with the bond market.

(iii) Precautionary demand for money

(eg:- Unforeseen Medical Bill)